CAPSTONE · AI FOR GOOD · WEEK 3

Learning curves, *and AI as an analyst*

Why offshore wind got cheaper, how fast it may keep getting cheaper, and how to check an AI assistant that tells you the answer.

Dr. Jan Erik Meidell · online · 18:15–20:15 · keep the shared canvas open on a second screen, on your team's Week 3 block

TONIGHT

Two hours, two exercises

18:18	A guess in the chat
18:21	Two teams, two curves
18:29	Wright's Law: numbers first, then the letters
18:46	Is it learning, or just time? And can it stop?
18:59	Exercise 1 · break the curve
19:17	Three 2040 cases, and report §3
19:23	Break · 10 minutes
19:33	What an assistant does, and five ways it fails
19:48	Exercise 2 · the audit and your AI policy
20:03	Vinuesa: AI helps, and AI costs · close

A GUESS IN THE CHAT · A PERCENTAGE

By how much did the cost of offshore wind power fall between 2010 and 2024?

About 61%: from about $203 to about $79 per MWh, global average for new farms.

Over the same years the world's offshore capacity went from about 3 GW to about 83 GW: more than four doublings.

Figures: the fitter's default rows on the Week 3 page, rounded from IRENA (cost) and GWEC (capacity) — the data lead verifies them. The percentage is computed.

TWO TEAMS, TWO CURVES · ON THE CANVAS

Read the other team's box first

Team 1

Rows used · fitted learning rate · R^2 · 2040 at the fitted rate and at 15% · the 2024 cost chosen, $79 or $86.

Team 2

The same five things. Same data, same tool.

If the two numbers differ, the difference is a decision someone made. Find it.

Canvas, Week 3 block, box "The team's curve".

WHERE A DIFFERENCE COMES FROM

Four decisions that move a learning rate

1. **Which rows.** Delete 2010 and the default fit goes from 19.8% to 21.1%.
2. **Which 2024 cost.** $79 from IRENA's rows, or $86 from the briefing.
3. **Which capacity.** The world's, or Europe's.
4. **Which tool.** An assistant that "fits" in its head rather than computing.

Learning rates computed from the fitter's default rows on the Week 3 page, rounded from IRENA (cost) and GWEC (capacity) — the data lead verifies them.

1936: *the cost of airplanes*

Theodore Wright, an aeronautical engineer, noticed that each time the number of planes built doubled, the labour needed per plane fell by about the same share: roughly 20%.

Not each year. Each **doubling of everything built so far**.

Wright, T. P. (1936). Factors affecting the cost of airplanes. *Journal of the Aeronautical Sciences*, 3(4), 122–128.

NUMBERS FIRST · THE DOUBLING LADDER

Start at $79 and 83 GW. Take 15% off at every doubling.

CUMULATIVE CAPACITY	DOUBLINGS	COST AT 15% PER DOUBLING	STARTING FROM $86 INSTEAD
83 GW	**0**	**$79**	**$86**
166 GW	**1**	**$67.1**	**$73.1**
332 GW	**2**	**$57.1**	**$62.1**
664 GW	**3**	**$48.5**	**$52.8**

Each rung: multiply by 0.85. Three rungs: 0.85 × 0.85 × 0.85 = 0.61.

Computed. Starting points: the fitter's default rows on the Week 3 page, rounded from IRENA (cost) and GWEC (capacity) — the data lead verifies them; $86 is the Week 2 briefing's figure.

THEN THE LETTERS

The same ladder, for any capacity

$$\text{Cost}(Q) = \text{Cost}(Q_0) \times (Q / Q_0)^{-b}$$

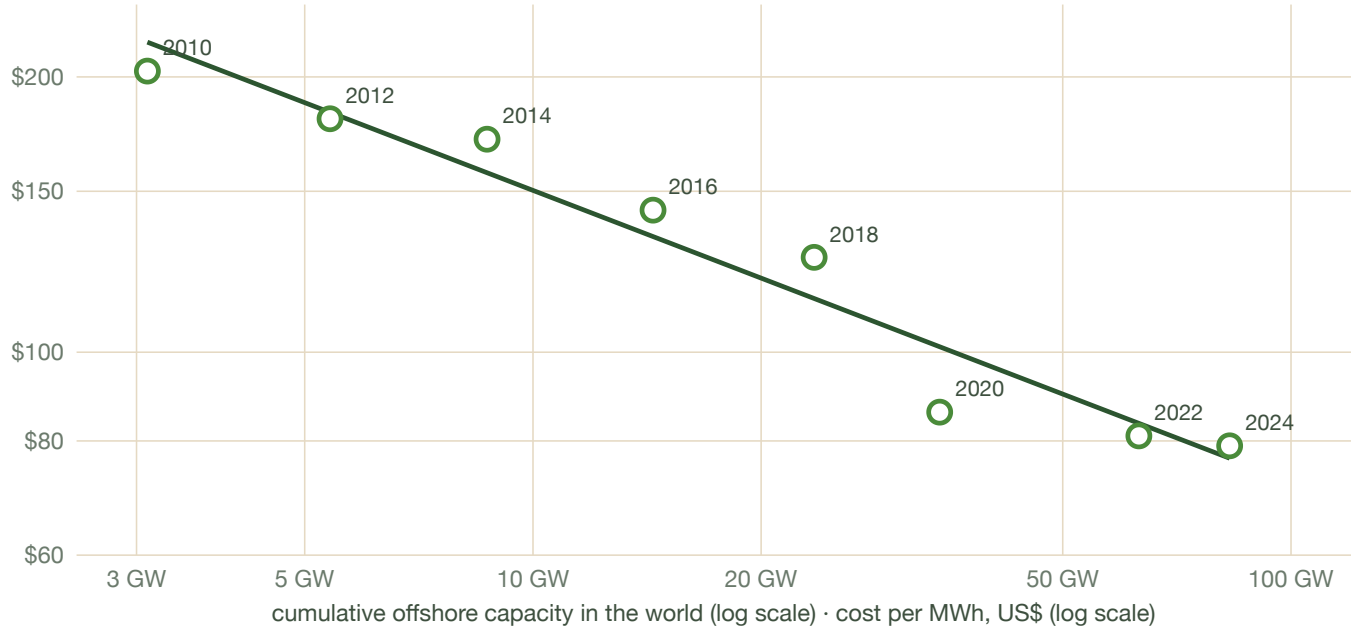
- **Q** · cumulative installed capacity, in GW
- **b** · how steep the curve is
- **learning rate = $1 - 2^{-b}$** · b = 0.234 gives 15%; b = 0.319 gives 19.8%

600 GW is 2.85 doublings beyond 83 GW, not three. That is why 2040 at 15% lands at about $49.7, not $48.5.

Computed: 600 / 83 = 7.23 = $2^{2.85}$.

WHY THE FITTER USES LOGARITHMS

On log scales, the curve becomes a straight line



Points: the fitter's default rows on the Week 3 page, rounded from IRENA (cost) and GWEC (capacity) — the data lead verifies them. Line: ordinary least squares of ln(cost) on ln(capacity); slope –0.319, learning rate 19.8%, R^2 0.94. Computed.

WHAT R^2 TELLS YOU · AND WHAT IT DOES NOT

0.94 says the line fits these eight points

It does say

94% of the variation in log-cost across these eight years lines up with log-capacity.

It does not say

That capacity *caused* the fall. That the line continues. That the rows are right. A wrong row that sits on the line raises R^2.

A high R^2 on eight points is a description, not a forecast.

WHOSE LEARNING?

The curve counts the world's gigawatts

Turbine makers, cable layers and installation vessels learn on every project, wherever it is built. A farm in China or Taiwan lowers the cost of one in Denmark.

The client is European. The learning is not. Week 2's tiles counted European gigawatts; the learning curve needs the global series.

Global cumulative offshore capacity, about 83 GW at the end of 2024: GWEC, Global Offshore Wind Report 2025.

IS IT LEARNING, OR JUST TIME?

Two laws that fit the past equally well

Wright

Cost falls with every doubling of cumulative production. More building, faster fall.

Moore

Cost falls by a fixed share every year, whatever happens.

Nagy and colleagues tested both on 62 technologies: both predicted about equally well, because production itself grew exponentially over time.

The difference matters for policy: under Wright, building more is what makes it cheaper.

Nagy, B., Farmer, J. D., Bui, Q. M., Trancik, J. E. (2013). Statistical basis for predicting technological progress. *PLoS ONE* 8(2): e52669.

WHY THIS MATTERS TO THE CLIENT

Trusting the curve has a price tag

Way and colleagues used Wright's Law to forecast energy costs. A fast transition to clean energy came out cheaper than no transition, by about $12 trillion in net present value.

Every serious outlook rests on a learning rate somebody chose. Yours is one of them.

Way, R., Ives, M. C., Mealy, P., Farmer, J. D. (2022). Empirically grounded technology forecasts and the energy transition. *Joule* 6(9), 2057–2082.

THREE TECHNOLOGIES, ONE PATTERN

The falls between 2010 and 2023

TECHNOLOGY	FALL IN THE AVERAGE COST OF NEW PROJECTS, 2010–2023
Utility-scale solar PV	**90%**
Onshore wind	**70%**
Offshore wind	**63%**

Offshore wind learns more slowly: every farm is a large construction project at sea, not a product off an assembly line.

IRENA (2024), *Renewable Power Generation Costs in 2023*, global weighted-average LCOE.

THE CURVE CAN BEND

2021–2024: offshore costs went up

- Steel, vessels and turbines became more expensive after 2021.
- Interest rates rose, and an offshore farm is mostly paid for before it produces anything.
- In the UK's 2023 auction (AR5), no offshore wind project bid at all: the price cap was too low.

Wright's Law counts gigawatts. It does not see interest rates. Week 4 does.

UK Department for Energy Security and Net Zero, Allocation Round 5 results, September 2023.

HOW MUCH THE FIT LEANS ON EACH ROW

Move one row's cost by ±20%, refit

ROW MOVED	COST −20%	COST +20%
2010	**17.5%**	**21.6%**
2016	**19.6%**	**20.0%**
2018	**20.2%**	**19.5%**
2024	**21.8%**	**18.1%**

From 19.8% with no row moved. The ends move the fit most; the middle barely moves it. That is why the data lead checks 2010 and 2024.

Computed from the fitter's default rows on the Week 3 page, rounded from IRENA (cost) and GWEC (capacity) — the data lead verifies them.

EXERCISE 1 · 14 MINUTES · BREAKOUT ROOMS

Break the curve

1. Each person takes one row nobody else has. Move its cost by ±20%; note the learning rate. Fill your line of **"The sensitivity test"**.
2. Then, together: find the **smallest honest change** to the rows that takes the rate below 15%. Honest means you could defend it to a client with a source.
3. One sentence in **"Which data points the fit is most sensitive to"**.

Use the fitter on the Week 3 page. Back at 19:13; each team reads its sentence.

READ-OUT

Two teams, one question

Which change took your rate below 15%, and would you put it in front of a client?

A good answer names the row, the source that justifies the change, and what it does to 2040.

THREE 2040 CASES

At 600 GW of offshore wind in the world

LEARNING RATE	FROM $79 IN 2024	FROM $86 IN 2024
Conservative · 10%	**$58.5**	**$63.7**
Central · 15%	**$49.7**	**$54.1**
Aggressive · 20%	**$41.8**	**$45.5**
The default fit · 19.8%	**$42.1**	**$45.8**

The choice of starting cost moves 2040 by about $4–6. The choice of learning rate moves it by about $17.

Computed. 600 GW in 2040 is the page's assumed global path, not a forecast.

A first paragraph, in four parts

1. The force: costs fall with every doubling of cumulative capacity.
2. The evidence: your rows, their sources, your fitted rate and R^2.
3. The limit: which rows the fit depends on, and that it does not see interest rates.
4. Why it is a trend, not a policy choice, and what it means for Ørsted and Eni.

Report §3, written by whoever the charter named.

BREAK

Ten minutes

Back at 19:33. Before you go: open the four assistants' answers to the audit question in the canvas box "The AI audit".

PART 2 · AI AS AN ANALYST

"Use AI" means four different things

METHOD	WHAT IT DOES	IN THIS CAPSTONE
Large language model	Writes text that is likely to follow your prompt	Drafting, summarising, structuring an argument
Regression	Finds the best line through data	Wright's Law, tonight
Time-series forecast	Extends past values forward	Capacity to 2040
Clustering	Groups similar cases	Mature and emerging wind markets

Asking a chat assistant for a learning rate is asking the first tool to do the second tool's job.

WHAT AN ASSISTANT DOES

It predicts the next word

A language model writes, one piece at a time, the text most likely to follow what came before. It was trained on text, not on a database of facts.

- A sentence that *sounds* like a sourced number is easy for it to produce.
- Whether the number is true depends on whether it appeared, often enough, in what it was trained on.
- Assistants that search the web cite pages; they still summarise them, and can misread them.

FIVE WAYS IT GETS IT WRONG

The audit table

FAILURE	WHAT IT LOOKS LIKE	HOW TO CATCH IT
Hallucinated data	A confident, specific number that does not exist	Primary source
Outdated knowledge	2022 figures presented as current	Ask for the year
Plausible maths error	GW read as GWh, a wrong capacity factor	Do it by hand
Out-of-distribution	A niche question, a confident wrong answer	Distrust perfect confidence
Confirmation bias	It agrees with how you framed the question	Ask it to argue the opposite

The same table as the Week 3 page, section 5.

A REAL CASE

Six court cases that did not exist

In 2023 two New York lawyers filed a brief citing six earlier court decisions. ChatGPT had produced them; none existed. The court fined the lawyers and their firm $5,000.

When asked, the assistant had confirmed that the cases were real.

Mata v. Avianca, Inc., US District Court, Southern District of New York, opinion and sanctions order of 22 June 2023.

THE AUDIT QUESTION

What did Ørsted announce in 2023?

What the company reported

Impairments of DKK 28.4 bn in the first nine months of 2023, DKK 19.9 bn of it on Ocean Wind 1. For the full year: DKK 26.8 bn of impairments plus a DKK 9.6 bn provision for cancelling Ocean Wind 1 and 2.

Typical assistant errors

A rounded "$4 bn" with no period · nine-month and full-year figures mixed · Ocean Wind 1 and 2 merged · Hornsea or Revolution Wind named as cancelled · links to pages that do not exist.

Ørsted, Interim report Q3 2023 and Annual report 2023. Check the full-year figures in the annual report itself before you quote them.

SORTING · TYPE THE NUMBER

Which failure mode is it?

1. "In 2023 Ørsted cancelled Hornsea 3", with a link that does not open.
   1 · hallucinated: Hornsea 3 went ahead; Ocean Wind 1 and 2 were cancelled
2. Ørsted's 2021 strategy targets presented as current. **2 · outdated knowledge**
3. "A 1 GW farm produces about 1 TWh a year." **3 · maths: 1 GW × 50% × 8,760 h ≈ 4.4 TWh**
4. "Yes, offshore wind is clearly the better bet", after you asked "why is offshore wind the better bet?"
   5 · confirmation bias

EXERCISE 2 · 12 MINUTES · BREAKOUT ROOMS

From the audit to a rule

1. Complete **"The AI audit"** table: for each assistant, true · wrong · unverifiable, and the failure mode.
2. Rank the four assistants for this kind of question. Is the ranking the same for the learning-curve question?
3. The coordinator writes the team's **one change to the AI policy** in the canvas box. Example: "No figure from an assistant goes into the report without a link we opened."

Back at 20:00; each team reads its rule.

READ-OUT

Two rules

Read your rule. The other team asks one question: what would this rule have caught in tonight's audit?

VINUESA ET AL. (2020)

AI can help with most of the goals, and harm a third of them

Enabler

134 of the 169 targets of the Sustainable Development Goals, **79%**.

Inhibitor

59 of the 169 targets, **35%**.

Many targets are on both lists. The authors warn that much of the evidence is weak and that the research tends to report the positives.

Vinuesa, R., Azizpour, H., Leite, I., et al. (2020). The role of artificial intelligence in achieving the Sustainable Development Goals. *Nature Communications* 11, 233. doi:10.1038/s41467-019-14108-y

FOR WIND ENERGY · SDG 7 AND 13

Enabler and inhibitor in the same industry

Enabler

Forecasting wind output for the grid · siting turbines · predicting failures before they happen · planning grid connections.

Inhibitor

Electricity and water for data centres · demand that competes with other loads for clean power · confident wrong analysis that misleads decision-makers.

YOUR TWO COMPANIES

The same tool, pointed at different resources

Both Ørsted and Eni use data and AI. Eni also runs HPC6, among the most powerful computers in the world, used among other things to image rock formations underground in the search for oil and gas.

When your team compares their strategies, ask what each company points its computing at.

TOP500 list, November 2024 (HPC6, Eni, Ferrera Erbognone, Italy); Eni press releases on HPC6.

THE RECURSIVE QUESTION

What does your analysis cost?

Google reports that a median Gemini text prompt uses about 0.24 Wh and emits about 0.03 g of CO2e.

Eight people, 100 prompts each over the project: about 190 Wh and 24 g CO_2e by that measure. Small, but other models, longer prompts and images cost more, and companies measure differently. Weeks 6 and 7 come back to it, with the editor's AI log as your data.

Google, "Measuring the environmental impact of AI inference", August 2025. The class total is computed.

THIS WEEK

Where tonight goes

TONIGHT	WHERE
Your learning rate, three 2040 cases, the case you believe	**Forum 5** · learning curve
One thing an assistant got wrong, with the true answer and source	**Forum 6** · the AI audit
The fitted rate, the rows, the limits	**Report §3** · first paragraph
The rule from Exercise 2	**The charter** · AI policy

Forum 5 is due Saturday, Forum 6 on Sunday. Both stay open after the deadline.

CLOSE

A learning rate is *a decision, not a fact*

The curve is real, and it can still be bent by interest rates, steel and ships. The assistant is useful, and it can be confidently wrong. In both cases your job is the same: say which number you chose, and why.

Questions on the material: the forum of the week. Anything personal: the message panel, bottom right of every page.